12-28 第一篇

Minerals and Geology: The Earth Beneath Our Feet

Minerals are natural substances that make up rocks and the Earth's crust. They form under heat, pressure, and chemical changes over millions of years. Common minerals include quartz, feldspar, calcite, and pyrite. Geologists study how these minerals form, where they are found, and what they tell us about the history of our planet. Understanding minerals helps us find valuable resources and protect the environment. [1]

Rocks are made of one or more minerals. Igneous rocks form when magma or lava cools. Sedimentary rocks build up from sand, mud, or the remains of living things. Metamorphic rocks change when existing rocks are heated and squeezed deep underground. The rock cycle shows how these three types can turn into each other over time. This process has shaped mountains, valleys, and coastlines for billions of years. [2]

Minerals are used in almost everything we use daily. Copper carries electricity; iron and aluminium build cars and buildings. Rare minerals power our phones and computers. Mining brings these materials to the surface, but it can damage land and water if not done carefully. Sustainable mining and recycling help reduce the impact on the planet. [3]

Fossils often form in sedimentary rocks. They preserve traces of ancient life—dinosaurs, plants, and early humans. By studying fossils and the layers of rock around them, scientists can date events and understand how life evolved. Geology connects the past with the present and helps us prepare for the future, including climate change and natural disasters. [4]

Geology is more than a subject in school—it is the key to understanding our home. From the ground we walk on to the resources we depend on, the Earth's story is written in rocks and minerals. Whether we become geologists or not, knowing a little about the ground beneath us helps us make wiser choices for our planet. [5]

选项

A. In short, minerals are the building blocks of our world, and geology helps us understand and use them wisely. B. For these reasons, the rock cycle is central to understanding how the Earth's surface has changed over time. C. As a result, we must balance our need for minerals with care for the environment. D. These records show that geology is not only about rocks—it is about the history of life itself. E. In other words, geology touches our lives in countless ways, and basic knowledge benefits us all. F. For example, diamonds form when carbon is exposed to extreme heat and pressure deep in the Earth.

答案键（Answer Key）

[1] A [2] B [3] C [4] D [5] E

深度解析

- **[1] A**：
  - **线索**：空前介绍矿物的定义、形成条件、常见种类、地质学家研究内容、理解矿物对资源与环境的意义，空后需总结矿物与世界的关系。A的"building blocks of our world"与前文呼应，"understand and use them wisely"与前文对应。
- **[2] B**：
  - **线索**：空前说岩石由矿物组成、岩浆岩沉积岩变质岩的形成方式、岩石循环及三者转化、塑造地貌数十亿年，空后需总结岩石循环的核心地位。B的"the rock cycle is central to understanding"与前文对应，"how the Earth's surface has changed over time"与前文呼应。
- **[3] C**：
  - **线索**：空前说矿物用途广泛——铜导电、铁铝用于建筑汽车、稀有矿物支撑电子设备、采矿有环境风险、可持续开采与回收可减损，空后需总结需求与环保的平衡。C的"balance our need for minerals with care for the environment"与前文对应。
- **[4] D**：
  - **线索**：空前说化石多见于沉积岩、保存古生物痕迹、科学家据此测年与理解演化、地质学连接过去与未来，空后需总结 geology 不仅是岩石更是生命史。D的"These records"指代化石，"geology is not only about rocks—it is about the history of life itself"与前文呼应。
- **[5] E**：
  - **线索**：空前说地质学不仅是学科更是理解家园的钥匙、从脚下土地到依赖的资源、地球故事写在岩石矿物中、了解一点对我们做选择有益，空后需总结地质学与生活的广泛联系。E的"touches our lives in countless ways"与前文对应，"basic knowledge benefits us all"与前文呼应。
- **干扰项分析**：
  - **F. For example, diamonds form when carbon is exposed to extreme heat and pressure...**：以For example开头，为具体矿物形成案例（钻石）。各空均为概括或过渡句；[2]易被误选因涉及矿物形成，但该空需总结岩石循环对地表变化的理解，而非单一矿物的形成机制。